

Evaluating the impact of Benson's relaxation technique on anxiety and delirium among coronary artery bypass graft surgery patients.

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Anxiety and delirium are prevalent complications in cardiac patients undergoing invasive procedures like coronary artery bypass surgery, with untreated symptoms potentially leading to enduring physical and psychological consequences. Benson's relaxation technique, a method of attention focusing, has demonstrated efficacy in alleviating various symptoms, although findings are conflicting. Thus, this study aimed to assess the influence of Benson's relaxation on anxiety and delirium severity in open-heart surgery candidates. Conducted in 2022 at Qaem Hospital in Mashhad, this randomized controlled trial involved 60 patients. The intervention group received Benson's relaxation training pre-surgery and practiced it twice daily for 72 h post-admission. Standard care was provided to controls. Anxiety and delirium were assessed using standardized scales before and after the intervention. The demographic characteristics of both groups were comparable with no significant differences. Benson's relaxation significantly reduced delirium scores on the first and second days post-surgery ($P < 0.001$). However, the difference in delirium scores between the groups was not significant on the third day ($P = 0.129$). While the intervention led to a significant reduction in anxiety within the intervention group, no significant difference in anxiety reduction was observed between the intervention and control groups ($P = 0.579$). Incorporating Benson's relaxation technique, which showed significant effects in reducing early postoperative delirium, into postoperative care plans for open-heart surgery patients warrants consideration. However, given the lack of a significant effect on anxiety reduction compared to routine care, further research is necessary to evaluate its long-term effectiveness and broader impact on psychological outcomes. This study was registered in the Iranian Registry of Clinical Trials (no. c IRCT20210903052365N1) on 2022-01-28.

The effectiveness of postoperative delirium prevention, diagnosis, and intervention protocol in patients monitored in the intensive care unit after cardiac surgery: a quasi-experimental study.

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The incidence of delirium is high in the intensive care unit (ICU) after cardiac surgery. The development of evidence-based care protocols for delirium management and training of nurses in this regard can ensure effective management of delirium. This quasi-experimental study aimed to assess the effectiveness of a postoperative delirium prevention, diagnosis, and intervention protocol in patients undergoing monitoring in the ICU after cardiac surgery. This study included 64 patients who underwent cardiac surgery and met the inclusion criteria, along with 14 nurses working in the ICU. Patients were divided into control ($n = 32$) and intervention ($n = 32$) groups. The study comprised three phases: determining the incidence of delirium in the control group and the delirium diagnosis status of the nurses; providing training to nurses on postoperative delirium prevention, diagnosis and intervention protocol; implementing a preliminary study of the protocol; and finally, implementing the protocol in the intervention group. Statistical significance was set at $p = 0.05$, significant agreement was observed in the intervention group (kappa: 1.00) ($p < 0.001$). The delirium diagnosis rate of the nurses was 14.3% in the control group and 100% in the intervention group, which was a significant difference. The incidence of delirium was 21.9% in the control group and 9.4% in the intervention group, although the difference was not significant. Postoperative delirium prevention, diagnosis, and intervention protocol effectively

enhance delirium diagnosis compliance among researchers and nurses and improve the accuracy of delirium diagnosis among postcardiac surgery ICU patients. The implementation of this protocol is recommended for delirium management in such patients. This study was retrospectively registered at Clinicaltrials.gov on 19.02.2024 (Clinical Trials ID: NCT06268119).

Effects of statins on delirium following cardiac surgery - evidence from literature.

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Delirium is a common complication after cardiac surgery, being associated with significant mortality and morbidity. The pathogenesis of postoperative delirium (POD) is complex and multifactorial, involving an interaction of multiple predisposing and precipitating factors. There are several hypothesis regarding the underlying mechanisms of POD, and the most recent emerging one involves neuroinflammation, which is exacerbated by the cardiopulmonary bypass-induced systemic inflammatory response. Experimental and clinical studies have recently documented improved perioperative central neural protection exerted by statins because of their anti-inflammatory, immunomodulatory, and antithrombotic properties. The present review will focused on the possible protective effect exerted by preoperative statin administration on delirium following cardiac surgery.

Preoperative transcranial Doppler and cerebral oximetry as predictors of delirium following valvular heart surgery: a case-control study.

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Delirium is a frequent and serious complication after cardiac surgery with cerebral hypoperfusion as one from the key pathophysiological mechanisms. Middle cerebral artery (MCA) mean blood flow velocity (MFV) measured by transcranial Doppler has been used as a marker of cerebral perfusion, and cerebral oximetry (rSO₂) value as a marker of its adequacy. This prospective observational trial examined the predictive value of MCA MFV and rSO₂, measured immediately before induction of anesthesia, for delirium after valvular heart surgery in elderly patients. In 113 patients, delirium was evaluated for 7 days postoperatively, using the confusion assessment method for the intensive care unit. The primary endpoint was the occurrence of postoperative delirium. Overall, 16 patients (14%) exhibited delirium. MCA MFV values could not predict the development of delirium. Preoperative statin use, geriatric depression scale score, and low preoperative rSO₂ (< 60%) showed association with delirium occurrence in univariable analysis. After multivariable analysis, only the low preoperative rSO₂ (< 60%) (OR 6.748, 95% CI 1.647-27.652, P = 0.008) remained as an independent predictor of delirium. Preoperative MCA MFV was not significantly associated with delirium after valvular heart surgery in elderly patients, while a low baseline rSO₂ value was associated with a sevenfold increased risk of delirium.

Treatment of delirium following cardiac surgery.

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Postoperative delirium is a common medical complication following cardiac surgery. This paper will

outline the treatment options for delirium with a focus on prophylactic use of risperidone before cardiac surgery.

Current approach to diagnosis and treatment of delirium after cardiac surgery.

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Delirium after cardiac surgery remains a common occurrence that results in significant short- and long-term morbidity and mortality. It continues to be underdiagnosed given its complex presentation and multifactorial etiology; however, its prevalence is increasing given the aging cardiac surgical population. This review highlights the perioperative risk factors, tools to assist in diagnosing delirium, and current pharmacological and nonpharmacological therapy options.

Cognitive Dysfunction, Delirium, and Stroke in Cardiac Surgery Patients.

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Neurologic injury in the form of cognitive decline, delirium, and stroke are common phenomena in patients undergoing cardiac surgery and continues to be one of the most common complication after cardiac surgery, in spite of improvements in mortality and improved surgical and anesthetic techniques. These complications lead to a significant increase in length of stay in the intensive care unit, increased length of hospital admission, and functional impairment, resulting in not only profound negative effects on patients who experience these complications, but also to increased costs of medical care and delivery. We discuss each of these complications in regard to their risks factors, incidence, potential therapeutic modalities, and relevant intraoperative and postoperative considerations.

Implementation of Evidence-Based International Recommendations Reduces Postoperative Delirium Rate in Patients Undergoing Cardiac Surgery or Interventions: A System-Based Quality Improvement Study.

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Delirium is a frequent and serious complication of cardiac procedures that can lead to serious long-term health restrictions. As primary prevention is more effective in reducing rate of delirium than the therapy itself, this study aimed to investigate the effect of a multidisciplinary delirium prevention bundle on the postoperative delirium rate in patients undergoing cardiac procedures. In this system-based quality improvement study, a four-component delirium prevention bundle was implemented in patients undergoing cardiac procedures at a single high-volume center. The program included preoperative delirium risk stratification, multidisciplinary education of consensus guidelines, written memory aids, and post-anesthetic visits with delirium screening until postoperative day three. Overall, 234 patients were included and analyzed during the 6-month study period. The overall delirium incidence rate was 12.4%. After the first 3-month baseline implementation period, the delirium rate was 17.2%, compared with 7.6% ($p = 0.026$) after implementation of the delirium prevention bundle. Multivariate analysis revealed independent risk factors such as age [adjusted odds ratio (OR): 1.046; 95% confidence interval (CI):

1.002-1.092; $p = 0.042$], double valve surgery [adjusted OR: 13.1; 95% CI: 3.240-52.974; $p < 0.0001$], and peripheral artery disease [adjusted OR: 8.131; 95% CI: 2.336-28.306; $p < 0.001$]. Hospital stay was significantly longer in patients with delirium [median 13 (12-19.5) versus 12 (11-14) days, $p = 0.009$]. This multidisciplinary system-based quality improvement study demonstrated a beneficial effect on the delirium rate after the implementation of a four-component delirium prevention bundle in patients undergoing cardiac surgery or intervention. Furthermore, multivariate analysis revealed important independent risk factors for delirium development. This might improve early risk stratification and strategies for this high-risk patient collective.

Perioperative psycho-educational intervention can reduce postoperative delirium in patients after cardiac surgery: a pilot study.

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Postoperative delirium after cardiac surgery is associated with many consequences such as poorer functional recovery, more frequent postoperative complications, higher mortality, increased length of hospital stay, and higher hospital costs. The aim of this study was to evaluate the efficacy of perioperative psycho-educational intervention in preventing postoperative delirium in post cardiac surgery patients. We conducted a comparative retrospective study between 49 patients who had received perioperative psycho-educational intervention and 46 patients who had received standard care. The primary outcome was the incidence of postoperative delirium. Secondary outcomes included length of ICU stay, and severity and duration of postoperative delirium among the patients who had developed delirium. The incidence of postoperative delirium was significantly lower in the intervention group than that in the control group (12.24% vs. 34.78%, $P = 0.009$). Among the patients who had developed postoperative delirium, there was no statistical difference between the two groups regarding secondary outcomes. Our results show that the patients who received perioperative psycho-educational intervention were associated with a lower incidence of postoperative delirium after cardiac surgery than those who received standard care. Clinicians would be able to implement this psycho-educational intervention as part of routine practice to reduce delirium.

Incidence of delirium after cardiac surgery: protocol for the DELIRIUM-CS Canada cross-sectional cohort study.

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Delirium is a recognized complication of cardiac surgery and is the focus of increasing attention owing to its negative effect on postoperative outcomes. However, little is known about the actual incidence of delirium following cardiac surgery, with published rates ranging widely, from 3%-78%. We describe the protocol for the DELIRIUM-CS Canada study, which will use validated and easily implementable bedside tools to determine the incidence of postoperative delirium in a contemporary cardiac surgery population. We hypothesize that delirium, identified through a systematic and standardized screening protocol, is a highly prevalent, though variable, condition following cardiac surgery. The DELIRIUM-CS Canada study is a multicentre cross-sectional cohort study. Over a 3-month period, all patients undergoing major cardiac surgical procedures at 10 participating centres will be screened for postoperative delirium by means of the Intensive Care Delirium Screening Checklist or the Confusion Assessment Method for the Intensive

Care Unit. Delirium screening will be conducted for 7 days following the date of surgery or until the initial discharge from the intensive care unit. In addition to reporting an overall rate of delirium, we will report unadjusted and adjusted incidence rates of delirium by institution and for the entire cohort. Risk adjustment will be performed with the use of multivariate regression modelling techniques. The results of this study will provide valuable insight into the true burden of delirium among patients having undergone a major cardiac surgical procedure in the current era. This is the first step in creating a multifaceted delirium prevention/treatment clinical pathway for patients undergoing cardiac surgery. Trial registration: ClinicalTrials.gov, no. NCT02206880.
